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CONNECTOR

Abstract

A connector includes: a housing; a terminal accommodated in the housing; and a fastening member configured to electrically connect the terminal by fastening. The housing includes insertion restrictors arranged in a vicinity of the terminal. The terminal includes protrusions configured to be in contact with the insertion restrictors at a normal arrangement position of the terminal with respect to the housing. The insertion restrictors are arranged at different positions with respect to an insertion direction of the terminal into the housing. The protrusions are arranged at different positions with respect to the insertion direction of the terminal into the housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is based on and claims the benefit of priority from the prior Japanese Patent Application No. 2024-026350, filed on Feb. 26, 2024, the entire contents of which are incorporated by reference herein.

TECHNICAL FIELD

[0002] The disclosure relates to a connector.

BACKGROUND

[0003] JP 2020-78971 A describes a connector including a housing, a bus bar as a terminal accommodated in the housing, and a bolt as a fastening member that electrically connects the bus bar by fastening. In this connector, an inverter is accommodated in the housing, and the bus bar is arranged in the inverter by accommodating the bus bar in the housing. The inverter and the bus bar are brought into contact with each other and electrically connected to each other by fastening the bolt.

SUMMARY

[0004] The connector is not provided with a structure that restricts a position for inserting the terminal into the housing. For this reason, the position for arranging the terminal with respect to the housing becomes unstable, and the position for arranging the terminal may be adjusted when the fastening member is fastened, and thus the assemblability is low. In addition, erroneous assembly in which the terminal is accommodated in the opposite direction with respect to the housing may occur.

[0005] The disclosure is directed to a connector capable of improving assemblability and preventing erroneous assembly of a terminal.

[0006] A connector in accordance with some embodiments includes: a housing; a terminal accommodated in the housing; and a fastening member configured to electrically connect the terminal by fastening. The housing includes insertion restrictors arranged in a vicinity of the terminal. The terminal includes protrusions configured to be in contact with the insertion restrictors at a normal arrangement position of the terminal with respect to the housing. The insertion restrictors are arranged at different positions with respect to an insertion direction of the terminal into the housing. The protrusions are arranged at different positions with respect to the insertion direction of the terminal into the housing.

[0007] According to the above configuration, the assemblability of the connector can be improved, and erroneous assembly of the terminal in the connector can be prevented.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view of a connector according to a first embodiment.

[0009] FIG. 2 is an exploded perspective view of the connector according to the first embodiment.

[0010] FIG. 3 is a cross-sectional view of the connector according to the first embodiment when a second terminal is arranged at a normal position.

[0011] FIG. 4 is a cross-sectional view of the connector according to the first embodiment when the second terminal is arranged at the normal position.

[0012] FIG. 5 is a cross-sectional view of the connector according to the first embodiment when the second terminal is not arranged at the normal position.

[0013] FIG. 6 is a cross-sectional view of the connector according to the first embodiment when the second terminal is not arranged at the normal position.

[0014] FIG. **7** is a perspective view of a first housing of the connector according to the first embodiment.

[0015] FIG. **8** is an exploded perspective view of the first housing of the connector according to the first embodiment.

[0016] FIG. **9** is a perspective view of a second housing of the connector according to the first embodiment.

[0017] FIG. **10** is an exploded perspective view of the second housing of the connector according to the first embodiment.

[0018] FIG. **11** is a perspective view of a cover of the connector according to the first embodiment.

[0019] FIG. **12** is an exploded perspective view of the cover of the connector according to the first embodiment.

[0020] FIG. **13** is a cross-sectional view of a connector according to a second embodiment when a second terminal is arranged at a normal position.

[0021] FIG. **14** is a perspective view of the connector according to the second embodiment in which a part of the connector is taken as a cross section when the second terminal is accommodated in a housing.

[0022] FIG. **15** is a perspective view of the connector according to the second embodiment in which a part of the connector is taken as a cross section when the second terminal is accommodated in the housing.

[0023] FIG. **16** is a perspective view of the connector according to the second embodiment in which a part of the connector is taken as a cross section when a front holder is assembled to the housing.

DETAILED DESCRIPTION

[0024] Hereinafter, connectors **1** and **101** according to embodiments will be described in detail with reference to the drawings. Note that dimensional ratios in the drawings are exaggerated for convenience of description, and may be different from the actual ratios.

First Embodiment

[0025] A first embodiment will be described with reference to FIGS. **1** to **12**.

[0026] As illustrated in FIG. **1**, a connector **1** according to the present embodiment is applied to, for example, an electric component **201** mounted on a vehicle. The connector **1** electrically connects, for example, the electric component **201** and other electric components (not illustrated) mounted on the vehicle, such as a power supply and a device. As illustrated in FIGS. **1** and **2**, in the electric component **201**, for example, a mating terminal (not illustrated) is accommodated inside a casing **203**. The casing **203** is provided with an arrangement portion **205** including a through hole which is formed through a wall where the mating terminal is arranged. The connector **1** is assembled to the arrangement portion **205**, and the electric component **201** and the connector **1** are electrically connected to each other.

[0027] As illustrated in FIGS. **1** to **12**, the connector **1** includes a housing **3**, a cover **5**, a first terminal **7**, and a second terminal **9** as a terminal.

[0028] As illustrated in FIGS. **1** to **10**, the housing **3** includes a first housing **11** and a second housing **13**.

[0029] As illustrated in FIGS. **1**, **2**, **7**, and **8**, the first housing **11** is formed of, for example, an insulating material such as a synthetic resin. The first housing **11** is formed in a tubular shape partially insertable into the arrangement portion **205** of the electric component **201**. On the outer circumference of the first housing **11**, a plurality of (two in the present embodiment) component fixers **15** protruding from the arrangement portion **205** of the electric component **201** to the outer diameter side are provided. A collar **19** into which a bolt **17** to be fastened to a screw hole **207** provided in the casing **203** of the electric component **201** is inserted is insert-molded in the component fixer **15**. On a surface of the component fixer **15** facing the casing **203** of the electric component **201**, a positioning pin **21** to be inserted into a positioning hole **209** provided in the

casing **203** protrudes toward the casing **203**. The component fixer **15** positions the first housing **11** with respect to the electric component **201** in a state where a part of the first housing **11** is inserted into the arrangement portion **205** of the electric component **201** by inserting the positioning pin **21** into the positioning hole **209**. The component fixer **15** fixes the first housing **11** to the electric component **201** by fastening the bolt **17** to the screw hole **207** via the collar **19**.

[0030] An inner housing **23** is accommodated in the first housing **11** from the electric component **201** side. The inner housing **23** is formed of, for example, an insulating material such as a synthetic resin. The inner housing **23** is held inside the first housing **11** via engaging portions (not illustrated) that engage with each other. The inner housing **23** accommodates, for example, a plurality of (two in the present embodiment) signal terminals **27** electrically connected to terminal portions of a plurality of (two in the present embodiment) signal lines **25** electrically connected to the electric component **201**. A terminal holder **29** that holds the signal terminal **27** therein is assembled to the inner housing **23**. Note that a locking lance that engages with the signal terminal **27** and holds the signal terminal **27** may be provided inside the inner housing **23**. A packing **31** is assembled in close contact with the outer circumference of the first housing **11** on the electric component **201** side. The packing **31** is held on the first housing **11** by a front holder **33** assembled to the first housing **11** via engagement portions (not illustrated) that engage with each other. The packing **31** is in close contact with the casing **203** of the electric component **201** in a state where the first housing **11** is assembled to the electric component **201**, and seals between the first housing **11** and the electric component **201**. A plurality of (two in the present embodiment) the first terminals **7** are accommodated in the first housing **11** from the side opposite to the electric component **201**.

[0031] As illustrated in FIGS. **1** to **6**, **9**, and **10**, the second housing **13** is formed of, for example, an insulating material such as a synthetic resin. The second housing **13** has an upper portion formed in a tubular shape into which a part of the first housing **11** can be fitted, and a lower portion formed in a tubular shape communicating with the upper portion and extending in a direction orthogonal to the direction of fitting between the first housing **11** and the second housing **13**. A plurality of (two in the present embodiment) the second terminals **9** are inserted into the lower portion of the second housing **13** from below, and the plurality of second terminals **9** are accommodated in the upper portion of the second housing **13**. On the outer circumference of the upper portion of the second housing **13**, a plurality of (two in the present embodiment) component fixers **35** protruding from the arrangement portion **205** of the electric component **201** to the outer diameter side are provided. A collar **39** into which a bolt **37** to be fastened to a screw hole **211** provided in the casing **203** of the electric component **201** is inserted is insert-molded in the component fixer **35**. The component fixer **35** fixes the second housing **13** to the electric component **201** by fastening the bolt **37** to the screw hole **211** via the collar **39** in a state where the first housing **11** and the second housing **13** are fitted to each other. Note that the first housing **11** and the second housing **13** may be provided with engagement portions that engage with each other to hold the fitted state between the first housing **11** and the second housing **13**.

[0032] A front packing **41** is assembled in close contact with the upper portion of the second housing **13** on the first housing **11** side. The front packing **41** is held on the second housing **13** by a front holder **43** as a holder assembled to the second housing **13** via engaging portions (not illustrated) that engage with each other. In the front holder **43**, a plurality of (two in the present embodiment) holding pins **45** protrude toward the second terminal **9** accommodated in the second housing **13**. The front packing **41** is in close contact with the first housing **11** in a state where the first housing **11** and the second housing **13** are fitted to each other, and seals between the first housing **11** and the second housing **13**. A rear packing **47** is assembled in close contact with the upper portion of the second housing **13** on the cover **5** side. The rear packing **47** is held on the second housing **13** by a rear holder **49** assembled to the second housing **13** via engagement portions (not illustrated) that engage with each other. The rear packing **47** is in close contact with the cover **5** in a state where the second housing **13** is covered with the cover **5**, and seals between

the second housing **13** and the cover **5**.

[0033] As illustrated in FIGS. **1**, **2**, **11**, and **12**, the cover **5** is formed of, for example, an insulating material such as a synthetic resin. The cover **5** covers the outer circumference of the upper portion of the second housing **13** on the side opposite to the electric component **201**, and is formed in a shape capable of closing the opening of the second housing **13**. On the outer circumference of the cover **5**, a plurality of (two in the present embodiment) component fixers **51** protruding from the arrangement portion **205** of the electric component **201** to the outer diameter side is provided at the same positions as the component fixers **35** of the second housing **13**. A collar **53** into which a bolt **37** to be fastened to a screw hole **211** provided in the casing **203** of the electric component **201** is inserted is insert-molded in the component fixer **51**. The component fixer **51** fixes the cover **5** to the electric component **201** by fastening the bolt **37** to the screw hole **211** via the collar **53** in a state where the cover **5** covers the second housing **13**. By providing the component fixers **51** at the same positions as the component fixers **35** of the second housing **13**, the cover **5** can be fixed by the same bolt **37**, and the number of components can be reduced. Note that the second housing **13** and the cover **5** may be provided with engagement portions that engage with each other to hold a state where the cover **5** covers the second housing **13**. The cover **5** accommodates a short terminal **55** that electrically connects a plurality of (two in the present embodiment) the signal terminals **27** accommodated in the first housing **11**, for example, by press fitting or the like. When the cover **5** is assembled to the electric component **201**, the short terminal **55** electrically connects the plurality of signal terminals **27**, and detects that the cover **5** is normally assembled to the electric component **201**.

[0034] As illustrated in FIGS. **7** and **8**, the first terminal **7** is formed of a conductive material, and is formed, for example, in a columnar shape. The plurality of (two in the present embodiment) the first terminals **7** are accommodated in the first housing **11**. The arrangement position of the first terminals **7** with respect to the first housing **11** is held, for example, by press fitting. Mating connectors **57** and terminal connectors **59** each formed by screw holes are provided on both end sides of the first terminals **7**. For example, a mating screw member (not illustrated) formed of a stud bolt is fastened to the mating connector **57**, and is integrally fixed with the first terminal **7**. The mating terminal of the electric component **201** is arranged on the mating screw member, and a mating nut (not illustrated) is fastened to the mating screw member. By fastening the mating nut to the mating screw member, the mating terminal and the first terminal **7** come into contact with each other, and the mating terminal and the first terminal **7** are electrically connected to each other.

[0035] For example, a screw member **63** as a fastening member **61** formed of a stud bolt is fastened to the terminal connector **59**, and the screw member **63** is integrally fixed to the first terminal **7**. By fixing the screw member **63** to the first terminal **7**, the screw member **63** can be handled as one member with the first terminal **7**, and the number of components can be reduced, whereby management and handling can be simplified. Note that the screw member **63** may be formed into one member continuous with the first terminal **7**, for example, by forming the end portion of the first terminal **7** on the terminal connector **59** side in a columnar shape, and forming the outer circumference of the columnar portion in a screw shape. The second terminal **9** is arranged on the screw member **63**.

[0036] As illustrated in FIGS. **3** to **6**, **9**, and **10**, the second terminal **9** is formed of a conductive material, and is formed, for example, in a plate shape. The plurality of (two in the present embodiment) second terminals **9** are accommodated in the second housing **13**. In the second terminal **9**, a pin hole **65** into which the holding pin **45** of the front holder **43** is inserted is provided through the second terminal **9**. By inserting the holding pin **45** into the pin hole **65**, the second terminal **9** is held at a normal arrangement position with respect to the second housing **13**. The plurality of second terminals **9** are electrically connected to, for example, terminal portions of a plurality of (two in the present embodiment) cables **67** electrically connected to other electric components (not illustrated) by crimping, welding, or the like. A rubber plug **69** is assembled in

close contact with the outer circumference of the cable **67** on the second terminal **9** side. The rubber plug **69** is held to the second housing **13** by a lower holder **71** assembled to the second housing **13** via engagement portions (not illustrated) that engage with each other. The lower holder **71** is divided into two and assembled to the cable **67** via engaging portions (not illustrated) that engage with each other. By dividing the lower holder **71**, the assembly position on the cable **67** can be freely determined, and the assemblability can be improved. The rubber plug **69** is in close contact with the second housing **13** in a state where the second terminal **9** is accommodated in the second housing **13**, and seals between the second housing **13** and the cable **67**.

[0037] The second terminal **9** is provided with an arrangement hole **73** formed through the second terminal **9** at a position corresponding to the screw member **63** of the first terminal **7** in a state where the first housing **11** and the second housing **13** are fitted to each other. A nut **75** is arranged in the arrangement hole **73** as the fastening member **61**, and the nut **75** is rotatably and integrally assembled with the second terminal **9** by, for example, caulking or the like. By rotatably and integrally assembling the nut **75** with the second terminal **9**, the nut **75** can be handled as one member with the second terminal **9**, and the number of components can be reduced, whereby management and handling can be simplified. By rotatably assembling the nut **75** to the second terminal **9**, the nut **75** can be fastened to the screw member **63** even when the screw member **63** is integrally provided with the first terminal **7**. The nut **75** is arranged with a gap between the nut **75** and the outer surface of the second terminal **9**, and can be inclined in a direction intersecting the axis of the screw member **63**. Therefore, when the second terminal **9** is arranged in an inclined manner, the nut **75** is inclined, so that the axis of the screw member **63** and the axis of the nut **75** can be arranged substantially on the same line, and the nut **75** can be stably fastened to the screw member **63**.

[0038] In the fastening member **61** including the screw member **63** and the nut **75**, the nut **75** is arranged at the end of the screw member **63** in a state where the first housing **11** and the second housing **13** are fitted to each other. At this time, when the screw member **63** and the nut **75** are in contact with each other, and the second terminal **9** is inclined in the direction intersecting the axis of the screw member **63**, the nut **75** is inclined in the direction intersecting the axis of the screw member **63**. Due to the inclination of the nut **75**, the axis of the screw member **63** and the axis of the nut **75** are arranged substantially on the same line, and the inclination of the second terminal **9** can be absorbed. By fastening the nut **75** to the screw member **63** in a state where the nut **75** is arranged on the screw member **63**, the first terminal **7** and the second terminal **9** come into contact with each other, and the first terminal **7** and the second terminal **9** are electrically connected to each other.

[0039] Here, in a case where the second terminal **9** is inserted too far into the second housing **13** or in a case where the second terminal **9** is not sufficiently inserted into the second housing **13**, the screw member **63** and the nut **75** as the fastening member **61** are not aligned, and the first terminal **7** and the second terminal **9** cannot be electrically connected to each other. In addition, there is a possibility that erroneous assembly in which the second terminal **9** is accommodated in the opposite direction with respect to the second housing **13** occurs. Therefore, the second housing **13** and the second terminal **9** are provided with an insertion restrictor **77** and a protrusion **79** for arranging the second terminal **9** at a normal position in the second housing **13**.

[0040] A plurality of (two in the present embodiment) the insertion restrictors **77** are provided inside the second housing **13** so as to be positioned on both sides of the second terminal **9** in the width direction. Each of the plurality of insertion restrictors **77** extends toward the side surface of the second terminal **9**. The plurality of insertion restrictors **77** are arranged at different positions with respect to the insertion direction of the second terminal **9** into the second housing **13**.

[0041] A plurality of (two in the present embodiment) the protrusions **79** are provided so as to protrude outward in the width direction from the side surfaces of the second terminal **9** on both sides in the width direction. Similarly to the plurality of insertion restrictors **77**, the plurality of

protrusions **79** are arranged at different positions with respect to the insertion direction of the second terminal **9** into the second housing **13**.

[0042] As illustrated in FIG. **3**, when the second terminal **9** is inserted into the second housing **13** in the normal direction, the second terminal **9** can be inserted into the second housing **13** until each of the plurality of protrusions **79** comes into contact with each of the plurality of insertion restrictors **77**. In a state where each of the plurality of protrusions **79** is in contact with each of the plurality of insertion restrictors **77**, the second terminal **9** is accommodated at a normal arrangement position in the second housing **13**. Therefore, the fastening member **61** can be fastened without adjusting the arrangement position of the second terminal **9**, and the assemblability can be improved. In a state where each of the plurality of protrusions **79** is in contact with each of the plurality of insertion restrictors **77**, the holding pin **45** of the front holder **43** can be inserted into the pin hole **65** of the second terminal **9**, as illustrated in FIG. **4**. Therefore, the front holder **43** can be assembled to the second housing **13**, and it is possible to detect that the second terminal **9** is arranged at a normal position.

[0043] As illustrated in FIG. **5**, when the second terminal **9** is inserted into the second housing **13** in the opposite direction, the protrusion **79** located on the front side in the insertion direction comes into contact with the insertion restrictor **77** located on the rear side in the insertion direction. In a state where the protrusion **79** located on the front side in the insertion direction is in contact with the insertion restrictor **77** located on the rear side in the insertion direction, the second terminal **9** is not accommodated at the normal arrangement position in the second housing **13**. Therefore, the fastening member **61** cannot be fastened, and it is possible to detect that the erroneous assembly in the second terminal **9** has occurred. In a state where the protrusion **79** located on the front side in the insertion direction is in contact with the insertion restrictor **77** located on the rear side in the insertion direction, the holding pin **45** of the front holder **43** cannot be inserted into the pin hole **65** of the second terminal **9**, as illustrated in FIG. **6**. Therefore, the front holder **43** cannot be assembled to the second housing **13**, and it is possible to detect that the erroneous assembly to the second terminal **9** has occurred.

[0044] As an example of such assembly of the connector **1**, first, the first housing **11**, the second housing **13**, and the cover **5** are each assembled.

[0045] In the assembling of the first housing **11**, first, the signal terminal **27** provided at the terminal portion of the signal line **25** is accommodated in the inner housing **23**, and the terminal holder **29** is assembled. Next, the inner housing **23** and the first terminal **7** are accommodated in the first housing **11**. Next, the screw member **63** is fastened to the first terminal **7**. The first terminal **7** to which the screw member **63** is fastened in advance may be accommodated in the first housing **11**. Then, the packing **31** is mounted on the first housing **11**, and the front holder **33** is assembled, thus completing the assembly of the first housing **11**.

[0046] In the assembling of the second housing **13**, first, the nut **75** is assembled to the second terminal **9**, and the second terminal **9** is assembled to the end portion of the cable **67** on which the rubber plug **69** is mounted. Next, the second terminal **9** is accommodated in the second housing **13**, and the lower holder **71** is assembled. At this time, when the second terminal **9** is in the normal direction, each of the plurality of protrusions **79** is in contact with each of the plurality of insertion restrictors **77**, and the second terminal **9** is accommodated at the normal position in the second housing **13**. On the other hand, when the second terminal **9** is in the opposite direction, the protrusions **79** on the front side in the insertion direction is in contact with the insertion restrictors **77** on the rear side in the insertion direction, and the second terminal **9** is not accommodated at the normal position in the second housing **13**. Then, the front packing **41** and the rear packing **47** are mounted on the second housing **13**, and the front holder **43** and the rear holder **49** are assembled, thus completing the assembly of the second housing **13**. At this time, when the second terminal **9** is arranged at a normal position, the holding pin **45** is inserted into the pin hole **65**, and the front holder **43** can be assembled to the second housing **13**. On the other hand, when the second terminal

9 is not arranged at the normal position, the holding pin **45** cannot be inserted into the pin hole **65**, and the front holder **43** cannot be assembled to the second housing **13**.

[0047] In the assembly of the cover **5**, the short terminal **55** is assembled to the cover **5**, and the assembly of the cover **5** is completed.

[0048] In the assembling of the connector **1**, first, the first housing **11** is arranged in the arrangement portion **205** of the electric component **201** so that the positioning pin **21** is inserted into the positioning hole **209**. Next, the bolt **17** is inserted into the collar **19** of the component fixer **15** of the first housing **11**, the bolt **17** is fastened to the screw hole **207**, and the first housing **11** is assembled to the electric component **201**. Next, the second housing **13** is arranged so that the first housing **11** is fitted inside the upper portion of the second housing **13**. Next, the nut **75** is fastened to the screw member **63** to electrically connect the first terminal **7** and the second terminal **9**. Next, the cover **5** is assembled to the second housing **13**. Then, the bolt **37** is inserted into the collar **39** of the component fixer **35** of the second housing **13** and the collar **53** of the component fixer **51** of the cover **5**, and the bolt **37** is fastened to the screw hole **211** to complete the assembly of the connector **1**.

[0049] Such a connector **1** includes the housing **3**, the second terminal **9** as a terminal accommodated in the housing **3**, and the fastening member **61** that electrically connects the second terminal **9** by fastening. The housing **3** is provided with insertion restrictors **77** arranged in the vicinity of the second terminal **9**. The second terminal **9** is provided with protrusions **79** that are in contact with the insertion restrictors **77** at a normal arrangement position of the second terminal **9** with respect to the housing **3**. A plurality of the insertion restrictors **77** are provided so that the insertion restrictors **77** are arranged at different positions with respect to the insertion direction of the second terminal **9** into the housing **3**. A plurality of the protrusions **79** are provided so that the protrusions **79** are arranged at different positions with respect to the insertion direction of the second terminal **9** into the housing **3**.

[0050] When the second terminal **9** is inserted into the second housing **13** in the normal direction, the second terminal **9** can be inserted into the second housing **13** until each of the plurality of protrusions **79** comes into contact with each of the plurality of insertion restrictors **77**. In a state where each of the plurality of protrusions **79** is in contact with each of the plurality of insertion restrictors **77**, the second terminal **9** is accommodated at a normal arrangement position in the second housing **13**. Therefore, the fastening member **61** can be fastened without adjusting the arrangement position of the second terminal **9**, and the assemblability can be improved. On the other hand, when the second terminal **9** is inserted into the second housing **13** in the opposite direction, the protrusions **79** located on the front side in the insertion direction comes into contact with the insertion restrictors **77** located on the rear side in the insertion direction. In a state where the protrusion **79** located on the front side in the insertion direction is in contact with the insertion restrictor **77** located on the rear side in the insertion direction, the second terminal **9** is not accommodated at the normal arrangement position in the second housing **13**. Therefore, the fastening member **61** cannot be fastened, and it is possible to detect that the erroneous assembly in the second terminal **9** has occurred.

[0051] Therefore, in such a connector **1**, assemblability can be improved, and the erroneous assembly of the second terminal **9** can be prevented.

[0052] The front holder **43** as a holder for holding the second terminal **9** in the housing **3** is assembled to the housing **3**. The second terminal is provided with the pin hole **65**. The front holder **43** is provided with the holding pin **45** to be inserted into the pin hole **65** at a normal arrangement position of the second terminal **9** with respect to the housing **3**.

[0053] When the second terminal **9** is arranged at a normal position, the holding pin **45** of the front holder **43** can be inserted into the pin hole **65** of the second terminal **9**. Therefore, the front holder **43** can be assembled to the housing **3**, and it is possible to detect that the second terminal **9** is arranged at a normal position. On the other hand, when the second terminal **9** is not arranged at a

normal position, the holding pin **45** of the front holder **43** cannot be inserted into the pin hole **65** of the second terminal **9**. Therefore, the front holder **43** cannot be assembled to the housing **3**, and it is possible to detect that the erroneous assembly in the second terminal **9** has occurred.

Second Embodiment

[0054] A second embodiment will be described with reference to FIGS. **13** to **16**.

[0055] In a connector **101** according to the present embodiment, a housing **3** is provided with a detachment restrictor **103** that can be engaged with a protrusion **79** in the detachment direction of a second terminal **9** from the housing **3** at a normal arrangement position of the second terminal **9** with respect to the housing **3**.

[0056] Note that the same configurations as those in the first embodiment will be denoted by the same symbols, and the description of the functions will be omitted as the first embodiment can be referred to regarding the functions. Since the configurations are similar, the obtained actions and effects are equivalent.

[0057] As illustrated in FIGS. **13** to **16**, a second housing **13** as the housing **3** is provided with the detachment restrictor **103**. A plurality of (two in the present embodiment) the detachment restrictors **103** are provided inside the second housing **13** so as to be positioned on both sides of the second terminal **9** in the width direction. The plurality of detachment restrictors **103** are arranged apart from the plurality of insertion restrictors **77** so as to have gaps between the detachment restrictors **103** and the insertion restrictors **77** in which the protrusions **79** can be arranged, at positions corresponding to the insertion restrictors **77**. The detachment restrictor **103** protrudes in a direction orthogonal to the insertion direction of the second terminal **9**. In the detachment restrictor **103**, an inclined surface **105** inclined upward toward the front side in the insertion direction is formed on the rear side in the insertion direction of the second terminal **9** into the second housing **13**. In the detachment restrictor **103**, a locking surface **107** serving as a vertical surface perpendicular to the insertion direction is formed on the front side in the insertion direction of the second terminal **9** into the second housing **13**.

[0058] When the second terminal **9** is inserted into the second housing **13**, the inclined surface **105** of the detachment restrictor **103** slides on the protrusion **79**, and the second terminal **9** is moved so as to surmount the detachment restrictor **103**. The second terminal **9** that has surmounted the detachment restrictor **103** is arranged at a normal position in the second housing **13**, and the protrusion **79** is arranged so as to be engageable with the locking surface **107** of the detachment restrictor **103**. When the protrusion **79** engages with the locking surface **107**, detachment of the second terminal **9** from the second housing **13** is prevented, and the second terminal **9** can be held at a normal arrangement position in the second housing **13**. Therefore, the fastening member **61** can be stably fastened, and electrical connection reliability can be maintained. The second terminal **9** can be held at a normal arrangement position in the second housing **13** even when the holding pin **45** of the front holder **43** is not inserted into the pin hole **65** of the second terminal **9**. Therefore, when the second terminal **9** cannot be detached from the second housing **13**, it is possible to detect that the second terminal **9** is arranged at a normal position. By the engagement between the protrusion **79** and the detachment restrictor **103**, for example, the holding pin **45** can be inserted into the pin hole **65** without involving the operation of the operator holding the second terminal **9**, and the assemblability of the front holder **43** to the second housing **13** can be improved.

[0059] In such a connector **101**, the housing **3** is provided with the detachment restrictor **103** that is engageable with the protrusion **79** in the detachment direction of the second terminal **9** from the housing **3** at a normal arrangement position of the second terminal **9** with respect to the housing **3**.

[0060] The engagement between the protrusion **79** and the detachment restrictor **103** allows the second terminal **9** to be held at a normal arrangement position in the housing **3**. Therefore, the fastening member **61** can be stably fastened, and electrical connection reliability can be maintained. When the second terminal **9** cannot be detached from the housing **3**, it is possible to detect that the second terminal **9** is arranged at a normal position.

[0061] In the above embodiment, the screw member **63** is provided integrally with the first terminal **7**, and the nut **75** is rotatably and integrally assembled with the second terminal **9**, but the present invention is not limited thereto. For example, the screw member **63** may be provided integrally with the second terminal **9**, and the nut **75** may be rotatably and integrally assembled with the first terminal **7**. Alternatively, the nut **75** may be integrally provided on one of the first terminal **7** and the second terminal **9**, and the screw member **63** may be rotatably and integrally assembled with the other terminal. The fastening member **61** includes the screw member **63** and the nut **75**, but is not limited thereto, and the fastening member **61** may only include the screw member **63**.

[0062] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A connector comprising: a housing; a terminal accommodated in the housing; and a fastening member configured to electrically connect the terminal by fastening, wherein the housing includes insertion restrictors arranged in a vicinity of the terminal, the terminal includes protrusions configured to be in contact with the insertion restrictors at a normal arrangement position of the terminal with respect to the housing, the insertion restrictors are arranged at different positions with respect to an insertion direction of the terminal into the housing, and the protrusions are arranged at different positions with respect to the insertion direction of the terminal into the housing.
 2. The connector according to claim 1, further comprising: a holder assembled to the housing and configured to hold the terminal in the housing, wherein the terminal includes a pin hole, and the holder includes a holding pin inserted into the pin hole at the normal arrangement position of the terminal with respect to the housing.
 3. The connector according to claim 1, wherein the housing includes a detachment restrictor engageable with one of the protrusions in a detachment direction of the terminal from the housing at the normal arrangement position of the terminal with respect to the housing.
 4. The connector according to claim 2, wherein the housing includes a detachment restrictor engageable with one of the protrusions in a detachment direction of the terminal from the housing at the normal arrangement position of the terminal with respect to the housing.
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